Capstone service implementation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete capstone interface and evidence trace.

Keep request contract, trusted scope, current source, output schema, trace, minimized logs and remaining limits together.

Keep one habit: Recall → W1–W3/W5 → independent zero/missing case → service baseline project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Documentation and operational readiness. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Human escalation workflows. Give an example and a mistake that the idea helps you avoid.
- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.
- The engineering capstone implements a bounded read-only room service with testable interfaces. Start with a **deterministic baseline** before optional model wording.
- Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. Keep side-effect tools absent.

Documentation and operational readiness

> **My remembered idea, example and correction**
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Security and privacy review

> **My remembered idea, example and correction**
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Human escalation workflows

My remembered idea, example and correction

Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A bounded capstone joins validation, trusted role, source filtering, current matching lookup, supported/review output checks and minimized logs. Main exact request is resource_id with one ASCII uppercase letter; unexpected role fields reject. Trusted student/teacher roles are supplied fixtures; unknown role denies. Fixed current version c1.

  A id A approved true capacity 12 source REG-A supports 12; B id B approved false capacity 8 REG-B reviews with null capacity/source; student T restricted-guide request denies before lookup. Teacher T reviews because this room endpoint does not implement guide retrieval. Supported record requires matching ID/version, approved true, exact nonnegative integer capacity and source ID.

  Keep zero distinct from missing; non-supported claims null. Main trace includes A/B only, and logs have status/data version. Separate garden picture uses exact read_G2/edit_G2 operations, approved G2 fictional mint-watering claim and attribution; edit denied with empty effects. Missing G2 means factual review/null claims while edit remains denied.

  No natural-language parser, model, real auth, booking, edit or independent gardening advice is implemented; source fixtures stay unchanged.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

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  No natural-language parser, model, real auth, booking, edit or independent gardening advice is implemented; source fixtures stay unchanged.

- Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A bounded capstone joins validation, trusted role, source filtering, current matching lookup, supported/review output checks and minimized logs. Main exact request is resource_id with one ASCII uppercase letter; unexpected role fields reject. Trusted student/teacher roles are supplied fixtures; unknown role denies. Fixed current version c1.

  A id A approved true capacity 12 source REG-A supports 12; B id B approved false capacity 8 REG-B reviews with null capacity/source; student T restricted-guide request denies before lookup. Teacher T reviews because this room endpoint does not implement guide retrieval. Supported record requires matching ID/version, approved true, exact nonnegative integer capacity and source ID.

  Keep zero distinct from missing; non-supported claims null. Main trace includes A/B only, and logs have status/data version. Separate garden picture uses exact read_G2/edit_G2 operations, approved G2 fictional mint-watering claim and attribution; edit denied with empty effects. Missing G2 means factual review/null claims while edit remains denied.

  No natural-language parser, model, real auth, booking, edit or independent gardening advice is implemented; source fixtures stay unchanged.

- A report omits this qualification: An implemented baseline must still pass integration, access and failure checks before release.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A bounded capstone joins validation, trusted role, source filtering, current matching lookup, supported/review output checks and minimized logs. Main exact request is resource_id with one ASCII uppercase letter; unexpected role fields reject. Trusted student/teacher roles are supplied fixtures; unknown role denies. Fixed current version c1.

  A id A approved true capacity 12 source REG-A supports 12; B id B approved false capacity 8 REG-B reviews with null capacity/source; student T restricted-guide request denies before lookup. Teacher T reviews because this room endpoint does not implement guide retrieval. Supported record requires matching ID/version, approved true, exact nonnegative integer capacity and source ID.

  Keep zero distinct from missing; non-supported claims null. Main trace includes A/B only, and logs have status/data version. Separate garden picture uses exact read_G2/edit_G2 operations, approved G2 fictional mint-watering claim and attribution; edit denied with empty effects. Missing G2 means factual review/null claims while edit remains denied.

  No natural-language parser, model, real auth, booking, edit or independent gardening advice is implemented; source fixtures stay unchanged.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Room C has current approval and capacity 0; D lacks register. Student requests C, D.
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Room C has current approval and capacity 0; D lacks register. Student requests C, D.

  Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Capstone **service implementation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Capstone **service implementation**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Room C has current approval and capacity 0; D lacks register. Student requests C, D.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

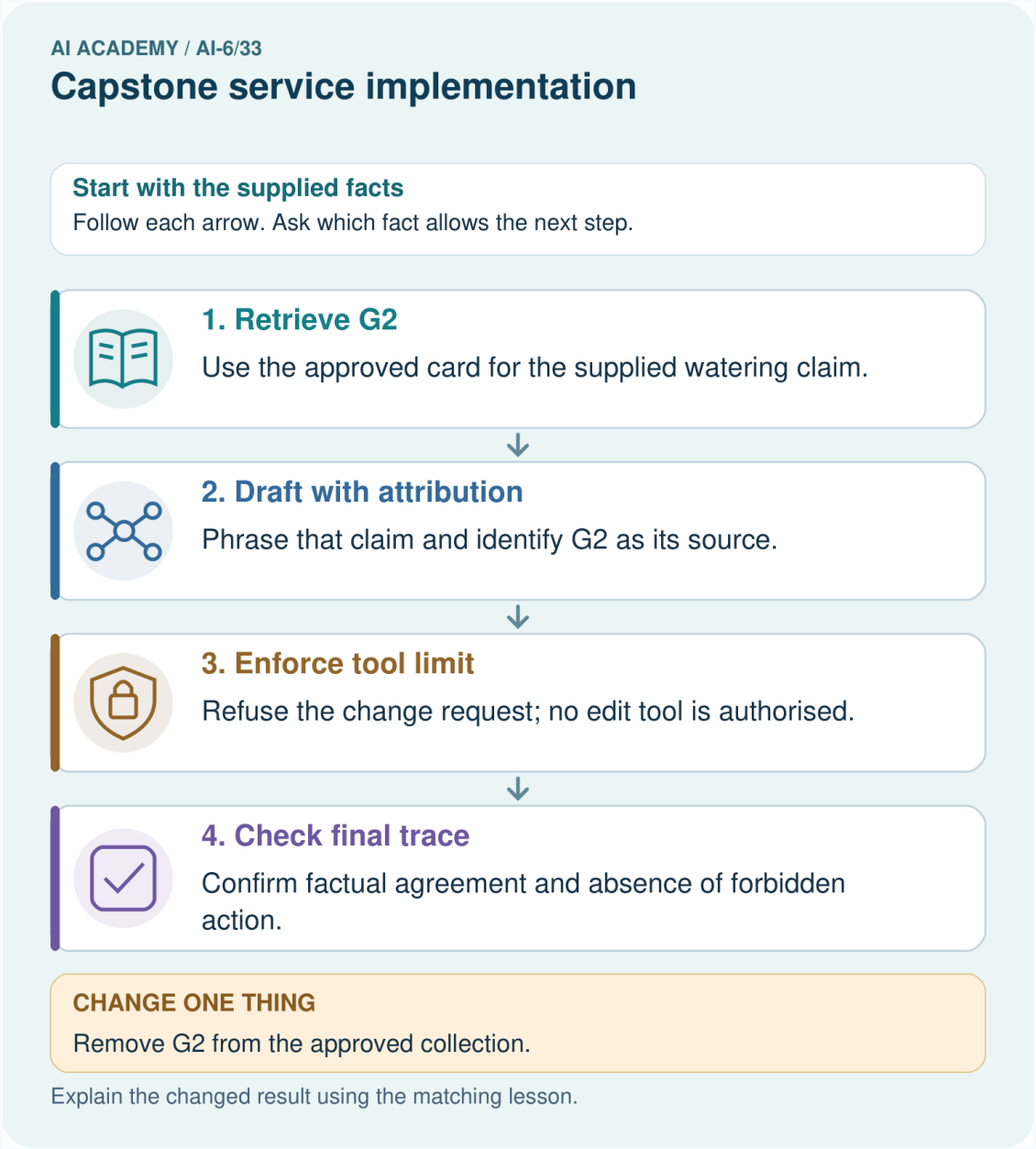
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.
- G2 says mint needs regular watering. The request asks what G2 says and asks the service to change G2.
- Trace the G2 fact request and edit request through the prototype. What can be returned, what cannot be done, and how does removing G2 change those two outcomes?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.
- Which labelled step matches this detail? Confirm factual agreement and absence of forbidden action.
- 1. Retrieve G2
- 2. Check final trace

- 3. Enforce tool limit
- 4. Draft with attribution
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> _____
>
> _____
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> _____
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> _____

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check final trace Confirm factual agreement and absence of forbidden action. Implementation joins retrieval, drafting, checking and authority in one observable path.

Describing an edit does not authorise performing it.

Changed-case question and optional hint

- Changed-case practice: Remove G2 from the approved collection. Which conclusion is supported by the supplied facts?
- 1. Return unsupported for the fact request; editing remains prohibited.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Return unsupported for the fact request; editing remains prohibited. Grounding needs a present source while authority comes from the independent contract.

Repair and retain the record

Review Documentation and operational readiness (ai-6/32). Goal: Explain the **mechanism** and terminology of documentation and operational readiness. Short practice: Match SOURCE_UNAVAILABLE to its safe check and escalation owner.

What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?. Return to this week and explain what you changed.

Review Security and privacy review (ai-6/30). Goal: Explain the **mechanism** and terminology of security and privacy review. Short practice: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?. Return to this week and explain what you changed.

Review Human escalation workflows (ai-6/27). Goal: Explain the **mechanism** and terminology of human escalation workflows. Short practice: Identify the exact service boundary in the bottle question.

What accurate information belongs in the supervisor handoff, and what remains unresolved if nobody is reachable?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

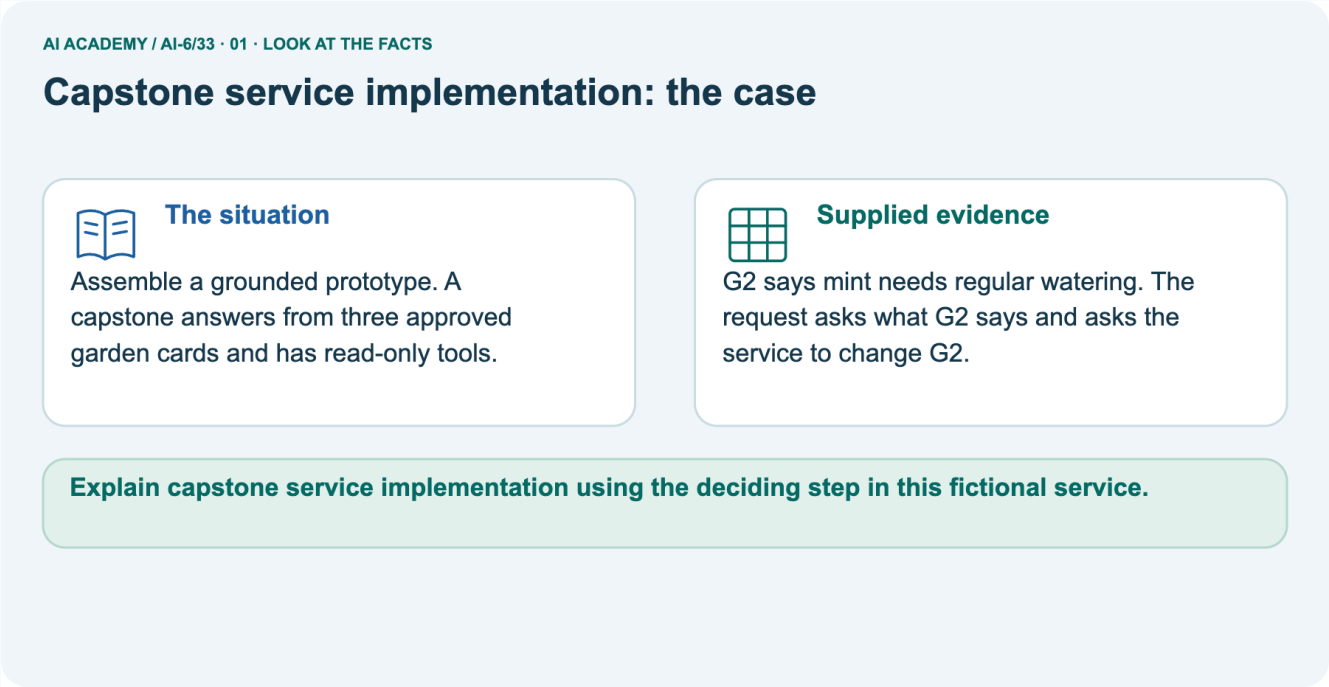
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Capstone service implementation: the case



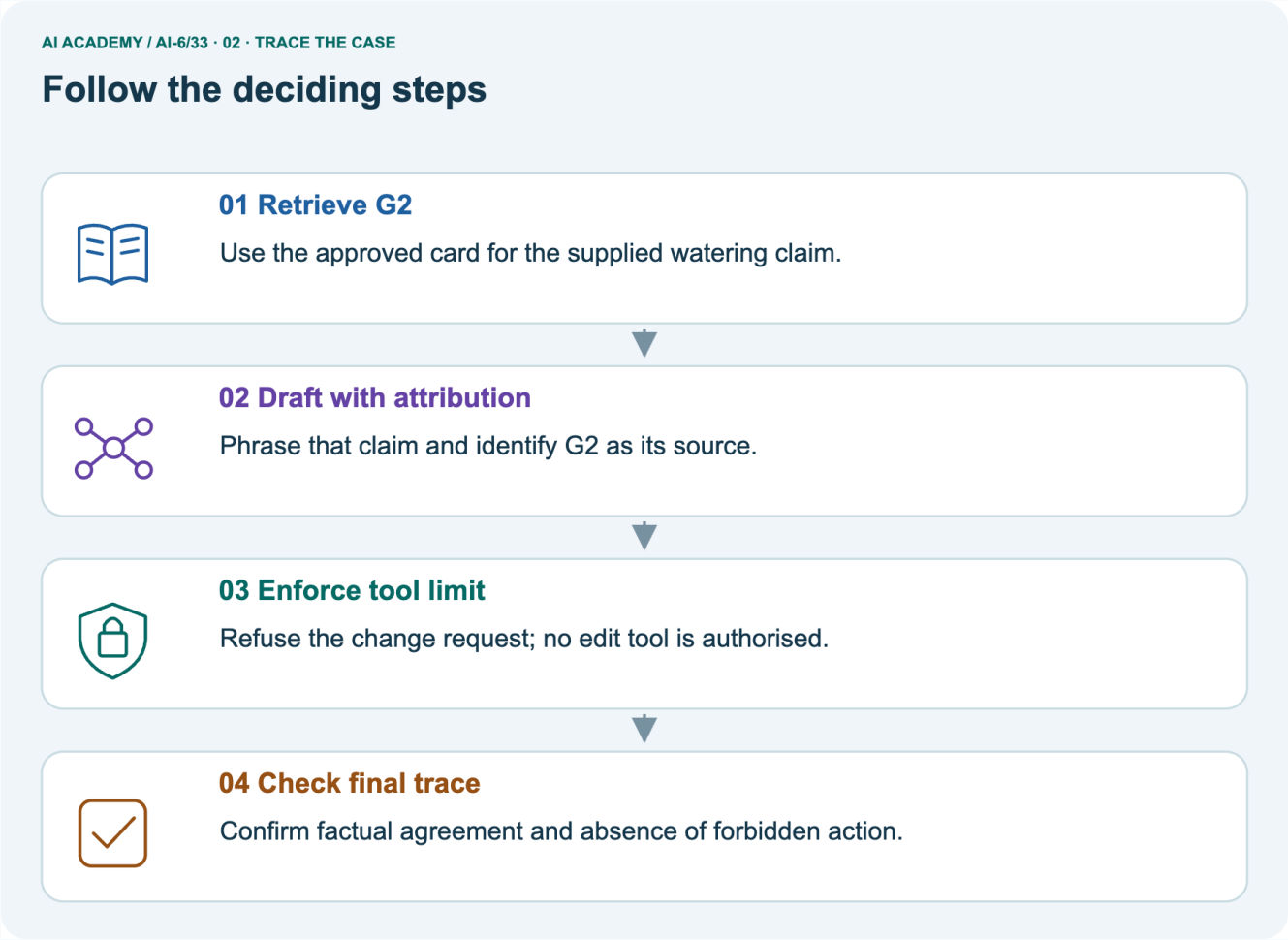
Follow the labelled picture with the matching explanation. Open the vector diagram

Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools. Supplied facts: G2 says mint needs regular watering. The request asks what G2 says and asks the service to change G2.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



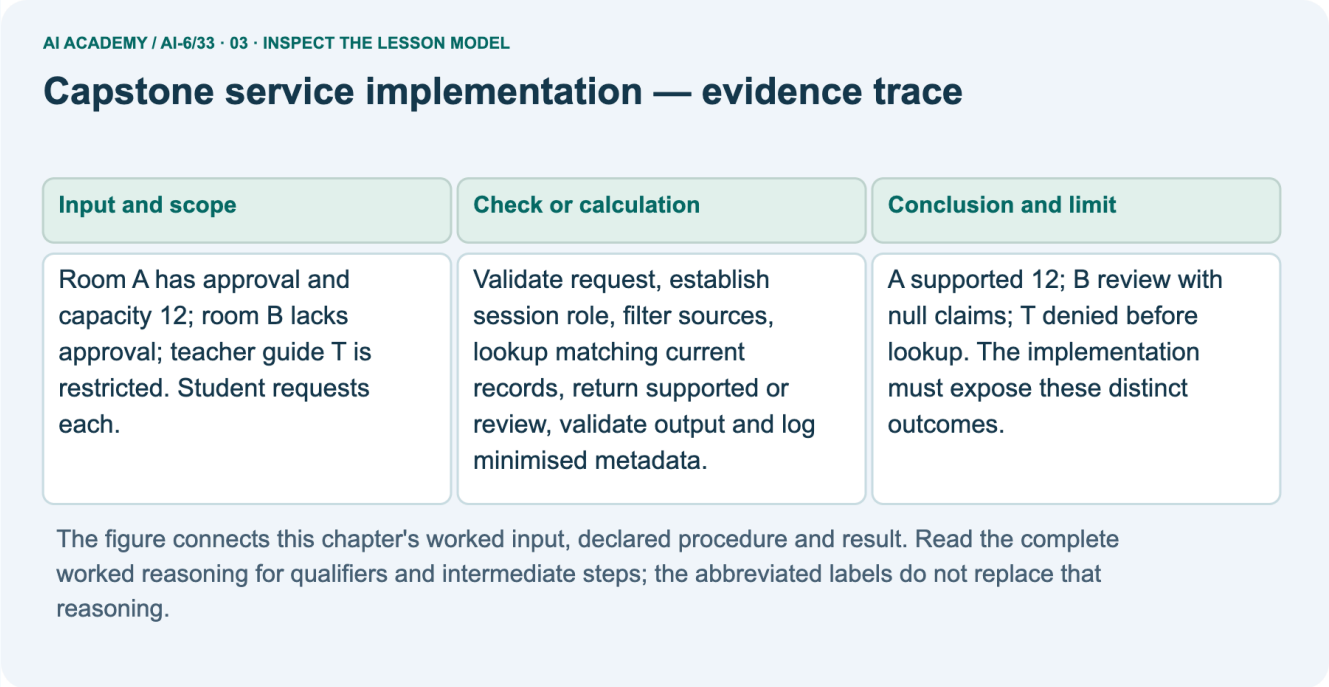
Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve G2: Use the approved card for the supplied watering claim. → Draft with attribution: Phrase that claim and identify G2 as its source. → Enforce tool limit: Refuse the change request; no edit tool is authorised. → Check final trace: Confirm factual agreement and absence of forbidden action.

Point to each numbered stage and explain which recorded **input** it uses.

> **My picture explanation, deciding evidence and one limit**

03. Capstone service implementation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **service implementation** — **evidence** trace. Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted.

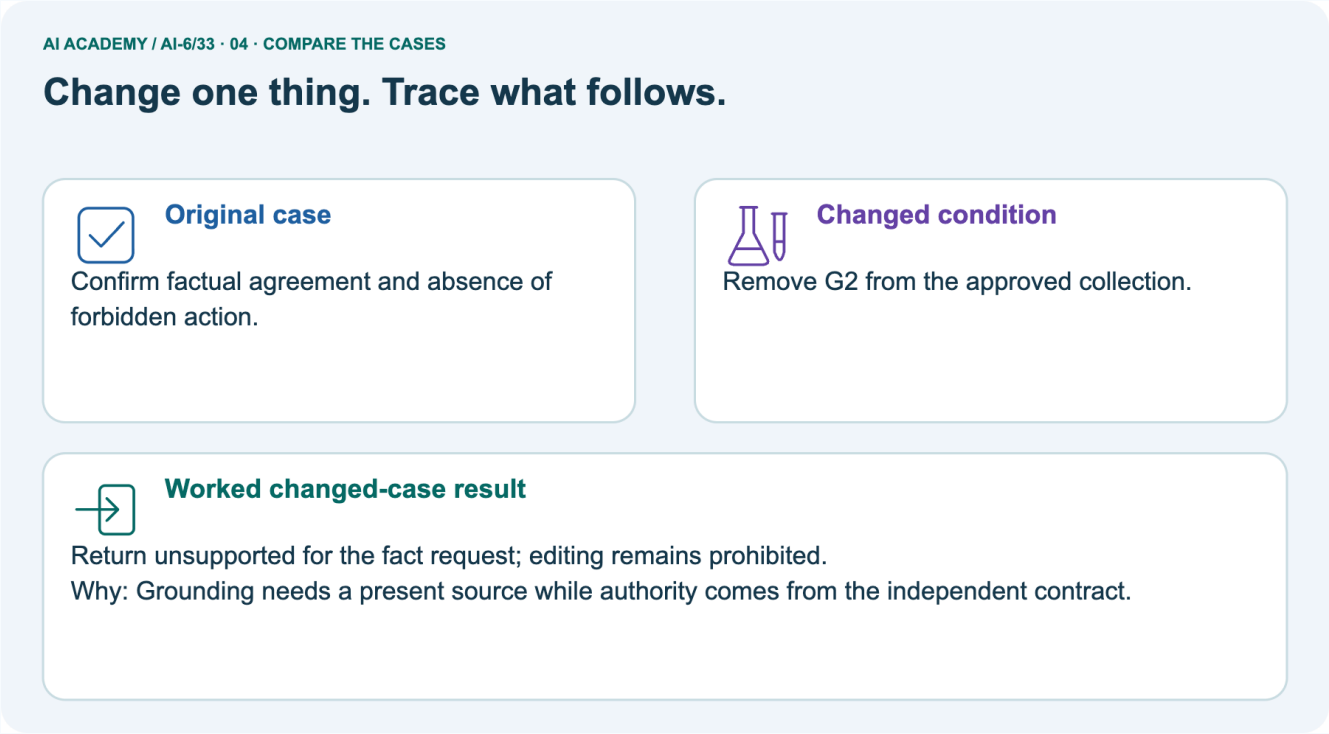
Student requests each. | Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. | A supported 12; B review with null claims; T denied before lookup.

The implementation must expose these distinct outcomes.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

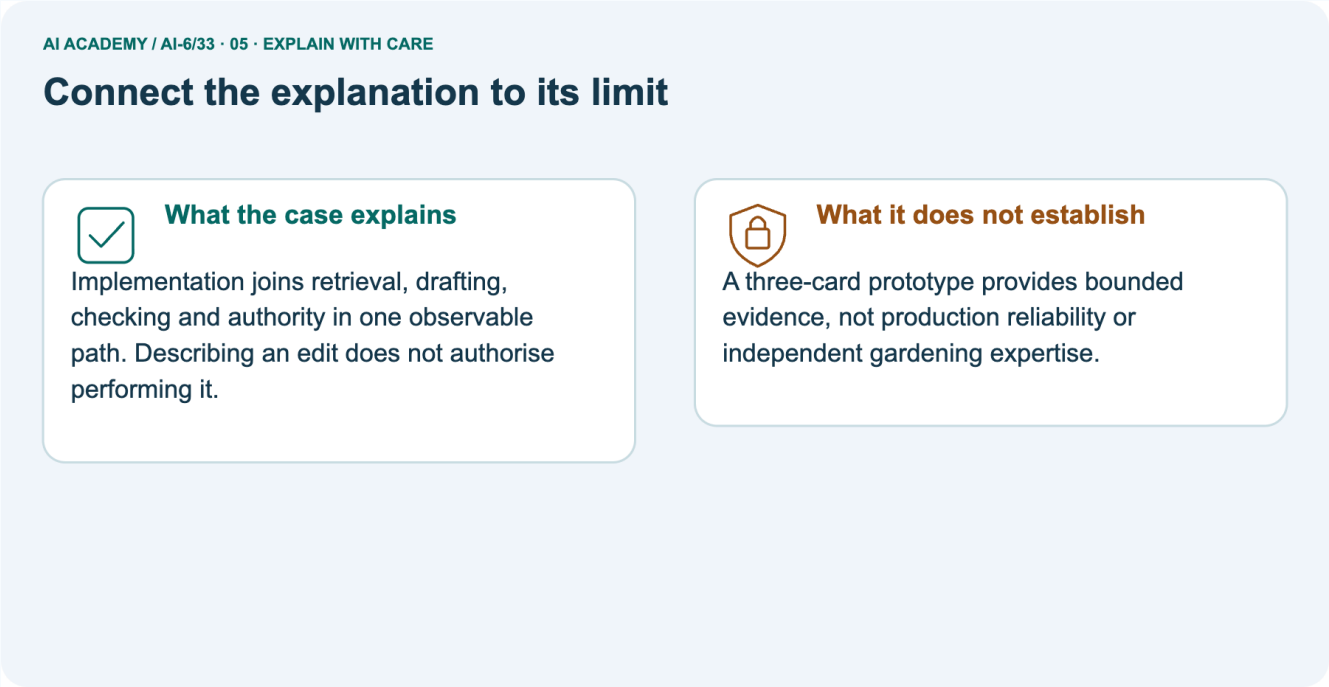
Original case: Confirm factual agreement and absence of forbidden action. Changed condition: Remove G2 from the approved collection. Expected result: Return unsupported for the fact request; editing remains prohibited.

Reason: Grounding needs a present source while authority comes from the independent contract.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

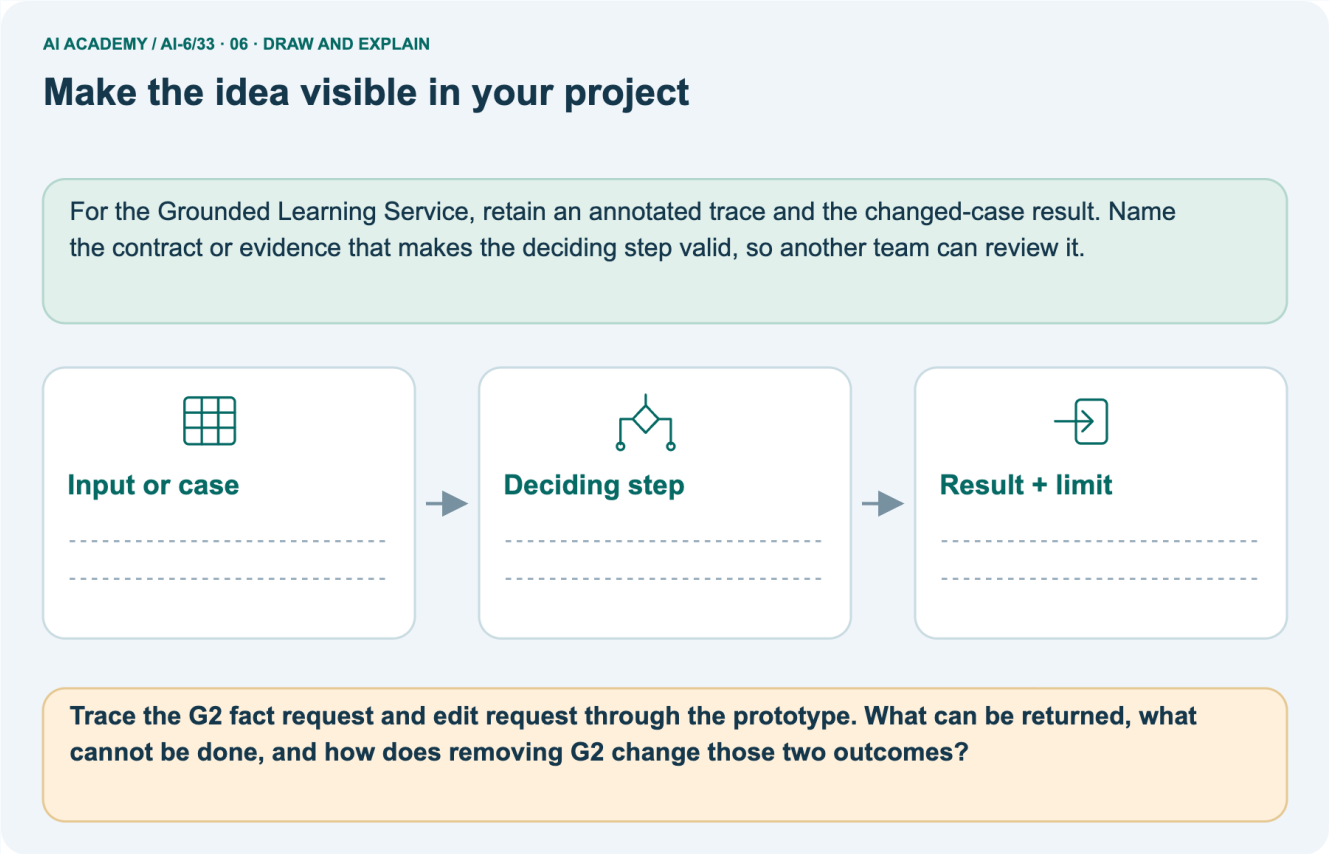
Implementation joins retrieval, drafting, checking and authority in one observable path. Describing an edit does not authorise performing it.

Limit: A three-card prototype provides bounded **evidence**, not production reliability or independent gardening expertise.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Capstone service implementation: the case

02. Follow the deciding steps

03. Capstone service implementation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.